

Binoy Saha

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EDUCATION

University of Texas at Arlington, Honors College

Aug 2025 - May 2029

Data Science (GPA: 3.8)

- **Coursework:** Calculus 1 & 2, Intermediate Programming, Python for Data Science 1 & 2, Data Mining

TECHNICAL SKILLS

- **Languages:** Python, Go, JavaScript, TypeScript, HTML/CSS, Java, SQL
- **Backend:** FastAPI, Node.js, Express.js, Spring Boot, REST API, GraphQL, JWT, OAuth2
- **Frontend:** React, React Native, Next.js
- **Databases:** PostgreSQL, MySQL, MongoDB, Redis, pgvector
- **DevOps & Cloud:** Docker, Kubernetes, CI/CD, AWS (EC2, S3), Linux, Unix, CI/CD, Nginx, Git
- **ML/AI:** PyTorch, YOLO, Scikit-learn, Whisper, Transformers, OpenCV, Apache Kafka, Apache Spark, NumPy, Pandas
- **Tools:** Git, Postman, VSCode, Agile / Scrum, Microservice Architecture

EXPERIENCE

ACM @ UT Arlington | *AI AutoReel - Backend Software Engineer*

Feb 2026

- Deployed a production-grade **REST API with JWT-based authentication middleware, bcrypt password hashing, CORS middleware, token-protected route guards, and structured error handling** (HTTP 401/404/500), consumed by a **React.js/Next.js** frontend serving a sports highlight generation platform
- Engineered a multi-stage video processing pipeline using **FastAPI** and **PostgreSQL**, orchestrating **FFmpeg** audio extraction, **OpenAI Whisper** **speech-to-text transcription**, **RMS/dB acoustic spike detection**, and **NLP keyword scoring into a single asynchronous job queue**, reducing highlight generation latency by **40%**
- Shipped a tiered subscription system (free/pro) with **per-user API quota enforcement, LLM-based highlight selection, and automated caption pipeline for pro users**, using **PostgreSQL** for usage tracking and session management
- Containerized full application stack using **Docker with nginx reverse proxy configuration**; managed collaborative development across a 6-person engineering team using **Git feature branching, pull requests, and code review**

State Farm, ACM UTA | *Undergraduate Research Assistant*

Feb 2026

- Built a **complete COCO-to-YOLO data preprocessing pipeline** handling **10,000+** annotated UAV images for automated hail and wind damage detection on rooftop imagery using object detection models
- Benchmarked **YOLOv8 (CNN-based) vs RF-DETR (transformer attention)** architectures across **precision, recall, and mAP** metrics to inform production deployment decisions for autonomous drone insurance systems; **co-authoring paper targeting ACM Multimedia Conference 2026**
- Containerized model training environment using **Docker** for reproducible deep learning experimentation across distributed team hardware configurations

Office of Information & Technology, UTA | *Frontend Developer*

Oct 2025

- Developed **25+ ServiceNow knowledge base pages** using **HTML, CSS, JavaScript, and Bootstrap** adhering to **WCAG 2.1 AA accessibility standards**, reducing IT ticket resolution time by **30% for 10,000+ campus users**
- Managed documentation version control using **Git, coordinating with 15+ department stakeholders** across distributed IT operations teams

RoboMasters Robotics, UTA | *Computer Vision Engineer*

Nov 2025

- Selected as only underclassman on competitive robotics team; **trained YOLOv8n object detection model on 100 CVAT-annotated frames extracted from game footage**, achieving **mAP50 of 0.939** and **armor plate recall of 0.982**
- Built **end-to-end computer vision pipeline** from raw video frame extraction and bounding box annotation in CVAT, to **COCO-to-YOLO data preprocessing, model training, and TensorRT export** for real-time inference on **NVIDIA Jetson Nano at RoboMasters North America 2026**

TECHNICAL PROJECTS

AI Agent Gateway | [Github](#)

Go, PostgreSQL, pgvector, Redis, Voyage AI, Anthropic Claude API

- Built **production AI agent gateway in Go** implementing **API key authentication, per-key rate limiting (10–100 req/min sliding window via Redis), prompt caching, and circuit breaker pattern** with half-open recovery - **6 core distributed systems infrastructure patterns implemented from scratch without frameworks**
- Reduced **repeated query latency by ~99% via Redis exact-match prompt cache (1hr TTL)**; engineered **agentic RAG pipeline** using **Voyage AI embeddings (1024-dim)** stored in **pgvector** with **sentence-boundary auto-chunking** and **HNSW cosine similarity search**, enabling LLM to autonomously decide when to retrieve context
- Implemented ensemble **self-reflection using 3 parallel Go goroutines** running **specialized critics (factuality, completeness, groundedness)** with majority vote evaluation - triggers automatic **re-generation with structured critic feedback on failure**, reducing LLM hallucination rate across tool-augmented responses

TechPulse – Neural RAG Media Intelligence Pipeline | [Github](#)

Python, FastAPI, Pinecone, Sentence-Transformers, Claude API

- Built an **end-to-end news ingestion and retrieval pipeline** using **NewsAPI** across 6 major outlets, **embedding 100+ articles into a Pinecone vector store (384-dim, cosine similarity)** with a **sentence-transformers encoding layer for semantic chunking and retrieval**
- Engineered a **two-stage ranking system** replacing **standard cosine-only RAG** with a **neural re-ranking layer** powered by **Meta's TRIBE v2 fMRI foundation model**, predicting cortical activation across **20,484 brain vertices** to score content by **human neural engagement**
- Designed and deployed a **FastAPI microservice** exposing the **TRIBE inference pipeline** as a **REST endpoint, enabling modular plug-in re-ranking on top of any retrieval system**; integrated **Claude Sonnet** downstream for multi-outlet framing analysis
- Validated **re-ranking signal against baseline cosine retrieval** across 6 outlets and **50+ retrieved chunks per query**, **surfacing measurable neural engagement divergence (+0.0672 vs -0.0338 on identical geopolitical queries)** - demonstrating **outlet framing bias undetectable by semantic similarity alone**